

BDA Quiz-20 2025

* Indicates required question

1. Email *

2. Roll Number *

3. Password *

Question

4. Suppose we are using the DGIM algorithm to estimate the number of 1's in suffixes of a sliding window of length 40. The current timestamp is 100, and we have the following buckets stored: 1 point

End Time	Size
100 -----	1
98 -----	1
95 -----	2
92 -----	2
87 -----	4
80 -----	8
65 -----	8

Note: Shown timestamps are absolute values, rather than modulo the window size, as DGIM would do.

Suppose that at times 101 through 105, 1's appear in the stream. Compute the set of buckets that would exist in the system at time 105. Then identify all the correct buckets. Buckets are represented by pairs (end-time, size).

Check all that apply.

- ☐ (80, 8)
- ☐ (95, 8)
- ☐ (98, 4)
- ☐ (103, 1)
- ☐ (100, 2)

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